

Trapped by Stability

A falsification framework for acquisition-driven radiomic phenotypes

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Abstract We do not offer another radiomic phenotype from breast MRI. We expose a failure mode that survives standard internal validation: the **stability trap**, in which a partition is reproducible under resampling, looks prognostic, and has no biology underneath it. Running a conventional unsupervised pipeline on a 922-patient multi-vendor breast-MRI cohort produced exactly the kind of result that invites over-interpretation — a stable two-cluster partition with enhancement- and vascularity-like feature differences and a recurrence-free survival split in HR+/HER2+ disease. Scanner identity and pulse-sequence basis were each strongly associated with that partition; molecular subtype and whole-cohort recurrence were not. Manufacturer residualization and ComBat both removed the survival association and preserved internal clusterability while replacing the patient partition entirely. We report this as a falsification, not a biomarker, and propose the minimum acquisition-aware checks a phenotype claim should have to survive.

! The result is a negative result

Everything below was produced by a pipeline that a reviewer would recognise as standard. It passes the checks such pipelines are usually asked to pass. It is still wrong. That is the point of the paper.

What survives, and what does not

Table 1: Claim status after acquisition-aware falsification. Every claim a reader would want to make from the raw discovery fails except the one about reproducibility.

Claim	Status	Evidence
The raw two-cluster partition is internally stable	 Supported	Mean bootstrap ARI 0.893 across $20 \times 80\%$ subsamples
Its separating features read as vascularity	 Supported	SER / PE enhancement-texture descriptors,
It separates recurrence-free survival in HR+/HER2+	 Unadjusted only	Log-rank $p = 0.010$; HR 4.76 $\rightarrow$ 2.48 ($p = 0.29$) once manufacturer enters
It is defined by molecular subtype	 Not supported	$\chi^2 p = 0.146$; η^2 on PC1 = 0.005
It is defined by acquisition	 Supported	Manufacturer $p = 8.0 \times 10^{-104}$; TR η^2 on PC1 = 0.755
It survives harmonization as the same phenotype	 Not supported	ARI vs raw labels = 0.056 (residualized), 0.065 (ComBat)
The post-harmonization structure meets bio-marker criteria	 Not supported	Strongest association is with tumour size, not outcome

The falsification chain

The contribution is a procedure, so it is worth stating as one. Each stage asks a question that a stability metric cannot answer, and each has a failing answer in this cohort.

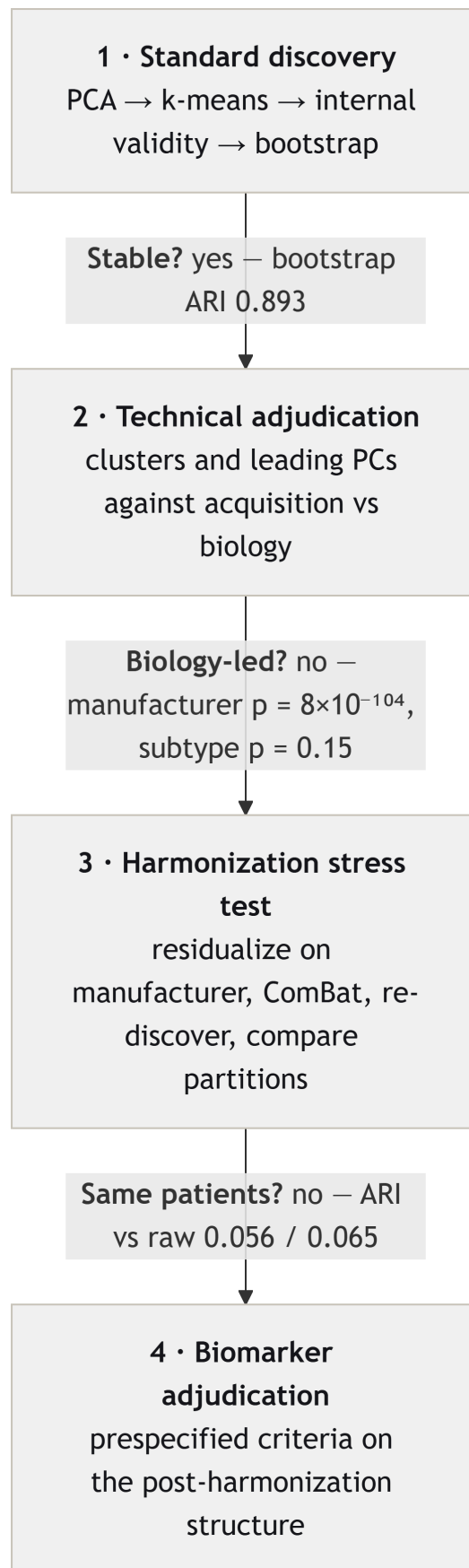


Figure 1: The acquisition-aware falsification chain. Each stage asks a question that a stabilifying metric cannot answer; the edge labels carry this cohort's answers. The claim survives stage 1 and fails every stage after it.

**Claim falsified —
downgrade to false positive**

⚠ The stability trap

A partition can be **reproducible** — the same solution recovered from resampled data — without being **valid** — the same patients grouped for the same reason. Bootstrap stability measures the first. Almost every unsupervised radiomics paper reports the first and concludes the second. When the dominant source of variance in a feature matrix is the scanner, the scanner’s structure is itself perfectly reproducible, and stability certifies it.

1 Introduction

Radiomics promises latent imaging subgroups — phenotypes invisible to standard labelling. Breast MRI and dynamic contrast-enhanced imaging yield quantitative features describing heterogeneity in tumour morphology, vascularity and cellularity, and unsupervised learning offers to find structure in them. In theory this lets clinicians recover stable, biologically meaningful groupings that go beyond receptor-based labels [1].

In practice, multi-vendor datasets have shown that acquisition variables — manufacturer, pulse-sequence parameters, field strength, voxel geometry, reconstruction choices, acquisition timing — can drive inference [2]. In high dimensions even small acquisition effects can dominate the feature space after dimensionality reduction, and a clustering algorithm will then recover a partition defined by technology rather than biology.

The validation practices in use measure reproducibility, not validity, and cannot distinguish a biologically meaningful phenotype from a technically induced one [3]. That gap is what we call the **stability trap**: the inference that a partition is biologically valid because the technical artifact underneath it is stable and reproducible.

This study therefore treats discovery and stability as the *beginning* of phenotype validation rather than its endpoint, and proposes a falsification chain that subjects any candidate phenotype to acquisition-aware challenges designed to expose technical confounding (Figure 1).

The first stage evaluates the raw feature space, testing whether the dominant principal components track biological or technical variables. The second stress-tests

Everything downstream follows from one asymmetry: stability is cheap to measure and identity is not. Only one of the two is routinely reported.

cluster identity: a real biological phenotype should survive manufacturer residualization and ComBat harmonization, so discovery is repeated after manufacturer effects are removed. The third re-evaluates the clinical interpretation after adjustment and across harmonization strategies.

Applying this to 922 breast-MRI patients, we first produce a stable two-cluster phenotype with enhancement-dominant, vascularity-like characteristics and an apparent HR+/HER2+ recurrence-free survival signal — and then show that the dominant variance axes are acquisition, not biology. The contribution is methodological rather than biomarker-driven: a phenotype can be internally stable, visually interpretable, and apparently prognostic while failing every test of biological identity.

2 Background and rationale

2.1 Radiomic phenotype discovery

Radiomics extracts quantitative information from routine images, promising knowledge of tumour phenotype not captured by visual interpretation, and positioning imaging as a bridge to personalized medicine alongside histology, genomics and clinical covariates [1]. Breast MRI is a particularly relevant modality: dynamic contrast-enhanced, diffusion-weighted and T2-weighted imaging together with enhancement kinetics carry information on vascularity, cellularity, morphology and tissue heterogeneity [4].

Most prior work is supervised, but the unsupervised setting is where phenotype discovery lives: patients are clustered on the feature space alone, and the resulting clusters are then examined for associations with prognosis, molecular subtype, immune state, stromal composition or pathway activity [5]. This is the route by which image-derived cancer subtypes have been proposed [6], [7], and there is broad agreement on the potential of radiomics to describe biological variation in tumours [8].

2.2 Supervised versus unsupervised settings

What separates the two is a failure mode that supervision partly avoids. With outcome labels that exist *outside* the feature space, model performance is judged against a predefined endpoint — molecular subtype, recurrence, treatment response — which limits the influence of technical confounding on inference [9]. This is much of why supervised radiomics has moved

toward clinical use faster than unsupervised radiomics has [10].

Unsupervised clustering has no such external anchor. It is built only from a feature space that may encode scanner, protocol and manufacturer effects. If technical differences explain more of the structure than tumour biology does, the algorithm recovers the hardware-defined partition first — and silhouette scores and bootstrap stability then test the reproducibility of that partition rather than its biological validity [3].

Mathematical reproducibility can be mistaken for biological validity.

2.3 Technical variance in radiomics

The literature on acquisition effects is extensive: features such as voxel size and gray-level discretization make many descriptors non-reproducible under acquisition variation [11], [12], [13], [14]. This is a large part of why radiomics has not fully transitioned into clinical workflow, despite standardization efforts such as the Image Biomarker Standardisation Initiative [2], [15]. MRI compounds the problem, since protocols, coils and pulse sequences differ across vendors [16].

The most direct proof of concept comes from multi-vendor breast-MRI radiomics: non-biological features predicted scanner platform and field strength with AUCs of 0.91–1.00, while the same pipeline predicted tumour biology only modestly (AUC 0.673 for pathological nodal status) without clinical variables [17]. Radiomic features can encode strong technical signal, and that signal competes with tumour biology inside the same feature space.

2.4 Harmonization and its limits

Feature-level harmonization is the standard remedy. ComBat, developed for batch effects in gene expression, was extended to neuroimaging and radiomics to remove scanner, site and protocol effects while retaining biological information [18], [19]. In radiomics it has been used almost exclusively to improve cross-site comparability or the transportability of supervised models [20], and it reduces batch effects and improves performance across PET, MRI and multi-centre settings [21].

But those evaluations assume the endpoint is what matters. That is the right frame for supervised prediction and the wrong one for unsupervised phenotypes, where the question is what happens to the *patient*

partition after harmonization. If a cluster reflects manufacturer variance more than tumour biology, ComBat can leave the mathematical stability intact while substituting a different partition that also passes validation. Standard metrics raise no alarm, because the mathematical properties have not changed – only the meaning has [22].

2.5 The validation gap

For prediction modelling it is accepted that internal performance is not enough: claims must face robustness checks, stress tests and external validation before generalizability is granted [23]. A phenotype claim should likewise be exposed to tests designed to *disprove* its biological interpretation. The question is not whether a reproducible cluster exists, but whether the same biological claim survives when technical variance is explicitly tested – whether clusters map more strongly to manufacturer than to subtype, whether the leading principal components are acquisition-dominated, whether identity and prognostic relevance survive residualization and ComBat, and whether what remains meets minimum biomarker criteria [24].

The argument is not that unsupervised radiomic discovery is invalid. It is that cluster stability is not equivalent to biological validity, particularly in multi-vendor data. The literature already establishes that feature space can be acquisition-dominated and that harmonization can mitigate it [6], [15], [21]; what is missing is an end-to-end falsification pipeline that tests whether a stable cluster retains its identity and interpretation once known acquisition effects are adjusted for.

3 Materials and methods

3.1 Data

We use a 922-patient breast-MRI dataset with 529 quantitative MRI-derived radiomic features [25], [26]. Variables were retained if they had <5% missingness and ≥ 2 unique values; under these criteria no features were removed. Missing values were median-imputed and all features standardized to zero mean before PCA, clustering, harmonization and survival modelling.

Z-scoring is not cosmetic here. The feature set mixes enhancement kinetics, peak enhancement, signal-enhancement ratio and texture statistics, all reported on different natural scales; without standardization, variance-driven methods such as PCA and k-means would be led by numerical scale rather than data structure.

Table 2: Radiomic matrix and preprocessing. No feature was lost to the filters — the 529-column matrix that enters PCA is the one the source data provides.

Step	Patients	Features	Result
Raw imaging matrix	922	529	Numeric MRI-derived radiomic features
Missingness filter (<5%)	922	529	No variable exceeded the threshold
Unique-value filter (≥ 2)	922	529	No constant variables present
Median imputation	922	529	Applied within retained variables
Z-score standardization	922	529	Zero mean, unit variance per feature
Matrix entering PCA	922	529	Input to PCA, k-means, harmonization

3.2 Standard phenotype discovery

PCA was applied to the standardized matrix and the leading components retained for visualization and clustering. Clustering used k-means across candidate k , with internal validity summarized by silhouette score, Calinski–Harabasz index and Davies–Bouldin index. Bootstrap stability was assessed by 20 repetitions of 80% subsampling and reported as the adjusted Rand index (ARI). Feature differences between clusters were assessed with Welch tests and Benjamini–Hochberg correction. This entire stage was run on the raw partition, to establish that it has a coherent imaging interpretation *before* any technical challenge is applied.

3.3 Falsification chain and technical adjudication

Four steps. Technical enrichment was measured by comparing cluster labels and PC structure against acquisition variables. Phenotype discovery was then repeated after manufacturer residualization and after ComBat harmonization. Partition identity was quantified by comparing harmonized labels with raw labels using ARI. Finally, clinical interpretability was retested after manufacturer adjustment.

3.4 Clinical and biomarker evaluation

The top screen was recurrence-free survival in the HR+/HER2+ subgroup, where the raw phenotype showed its strongest survival split, using Kaplan–Meier curves, the log-rank test and Cox proportional-hazards models. The cluster effect was then re-estimated with manufacturer in the model, to ask whether it remains interpretable after technical adjustment. The post-harmonization structure was finally assessed against prespecified biomarker criteria. Those criteria are deliberately minimal: they are identification thresholds, not confirmation of biological validity.

Steps 2 and 3 are the ones missing from conventional practice. Step 3 in particular has no standard reporting slot, which is why partition identity goes unreported even when it has collapsed.

4 Results

4.1 A standard workflow produces a plausible but fragile discovery

At first glance the output reads as a phenotype discovery.

The top ten principal components explain 57.9% of the variance, with PC1, PC2 and PC3 at 15.5%, 9.6% and 7.8% (Table 3). The scree plot and the PC1–PC2 projection (Figure 2, Figure 7) show a continuum rather than separated masses: any clustering here draws a gradient boundary, not a natural one.

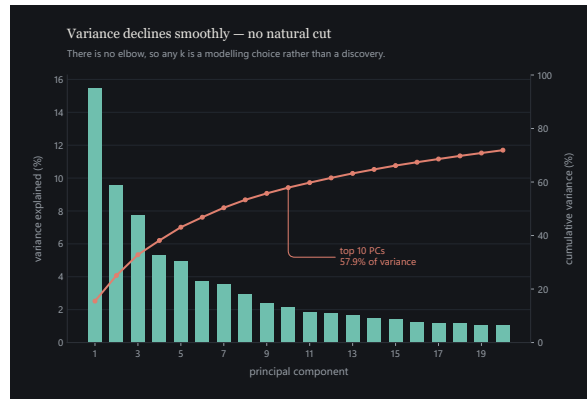
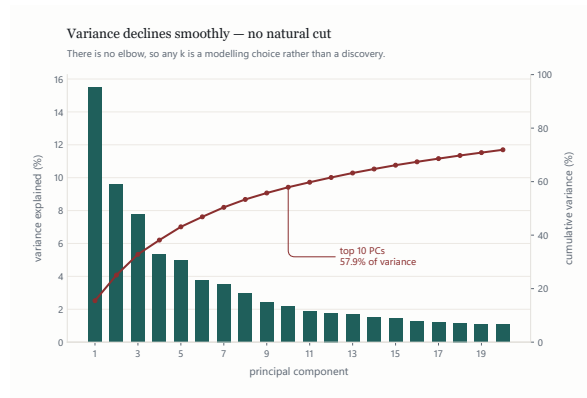


Figure 2: Scree plot and cumulative variance for the leading 20 principal components.

Table 3: Variance explained by the leading principal components.

Principal component	Variance explained	Cumulative
PC1	15.49%	15.5%
PC2	9.58%	25.1%
PC3	7.75%	32.8%
PC4	5.32%	38.2%
PC5	4.95%	43.1%
PC6	3.74%	46.8%
PC7	3.54%	50.4%
PC8	2.96%	53.3%
PC9	2.41%	55.8%
PC10	2.15%	57.9%

k-means selected $k = 2$ on the metrics that are conventionally reported: the highest silhouette score (0.196) and the highest Calinski–Harabasz index (200.9), both declining as k grows (Figure 3). Davies–Bouldin does not agree — $k = 2$ is its worst solution — which is

itself worth noticing, since the selection rests on two of three metrics and the disagreement is rarely reported.



Figure 3: Internal cluster validity across candidate k . Two of the three metrics select $k=2$; the third selects against it.

Bootstrap stability was strong: the raw $k=2$ partition attained a mean ARI of 0.893 (SD 0.084) across 20 subsamples (Figure 9). By the standards usually applied, this is where a phenotype paper would stop.

Feature-level inspection then supplied a biologically plausible reading. The top-ranked separating variables are dominated by signal-enhancement ratio (SER), peak enhancement (PE), entropy, homogeneity and related enhancement-texture descriptors, with $|\text{Cohen's } d|$ above 2.2 for the leading SER entropy and sum-entropy features (Figure 4, Table 4). The phenotype looks *vascularity-like*: one group with more spatially heterogeneous enhancement, the other more homogeneous. Whether that has a genuine biological basis is, for our purposes, beside the point — what matters is how readily the interpretation attaches.

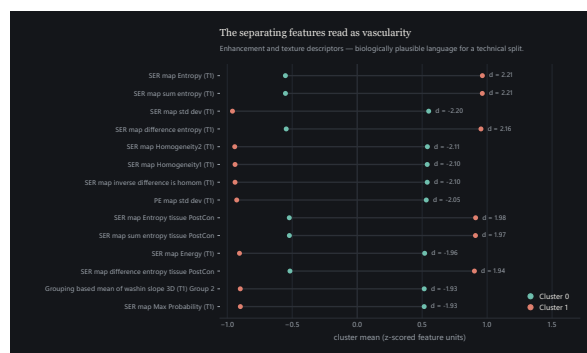
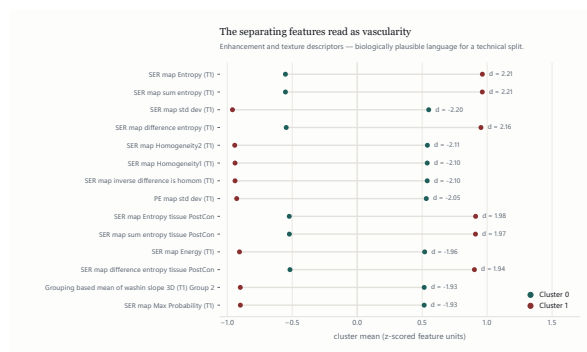


Figure 4: The 14 strongest cluster-separating features, as cluster means in z-scored units with Cohen's d.

Table 4: Top raw cluster-separating radiomic features
(Welch tests, Benjamini–Hochberg corrected).

Feature	Mean C0	Mean C1	Δ (C1–C0)	Cohen’s d	q- value
SER map En- tropy tissue T1	−0.552	0.964	+1.516	2.21	1.1×10^{-100}
SER map sum en- tropy tissue T1	−0.552	0.963	+1.515	2.21	2.7×10^{-109}
SER map std dev tissue T1	0.550	−0.960	−1.510	−2.20	2.5×10^{-164}
SER map dif- fer- ence en- tropy tissue T1	−0.546	0.953	+1.499	2.16	1.2×10^{-103}
SER map Homogeneity2 tissue T1	0.540	−0.942	−1.483	−2.11	2.5×10^{-95}
SER map Homogeneity1 tissue T1	0.539	−0.940	−1.479	−2.10	1.7×10^{-97}
SER map in- verse dif- fer- ence is ho- mom tissue T1	0.539	−0.940	−1.479	−2.10	1.7×10^{-97}
PE T1 map std dev tissue	0.532	−0.927 ¹³	−1.459	−2.05	1.7×10^{-164}

Finally, the raw partition looked prognostically meaningful in HR+/HER2+ disease. Of 101 patients with usable follow-up, Cluster 0 had 55 patients and 2 recurrences; Cluster 1 had 46 patients and 8. Kaplan–Meier curves separate visibly (Figure 5), the log-rank test is significant ($p = 0.010$), and an unadjusted Cox model gives $HR = 4.76$ for Cluster 1 versus Cluster 0 (95% CI 1.22–18.61; $p = 0.025$) (Table 5).

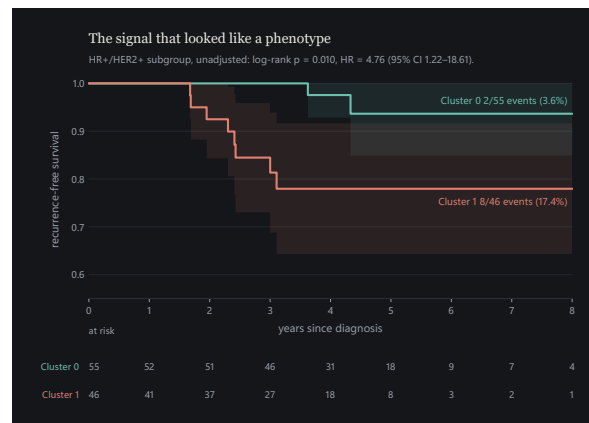
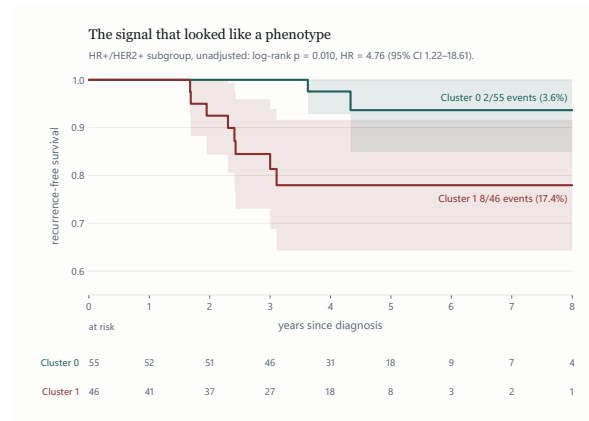


Figure 5: Recurrence-free survival in HR+/HER2+ patients by raw radiomic cluster, with 95% confidence bands and numbers at risk.

Table 5: The unadjusted HR+/HER2+ recurrence-free survival signal by raw radiomic cluster.

Analy- sis	Esti- mate	Inter- val / rate	p-value	Read- ing
Cluster 0	55 pa- tients, 2 events	3.6%	—	Lower unad- justed recur- rence burden
Cluster 1	46 pa- tients, 8 events	17.4%	—	Higher unad- justed recur- rence burden
Log- rank test	—	—	0.010	Curves sepa- rate
Cox, cluster only	HR = 4.76	95% CI 1.22– 18.61	0.025	Signif- icant before techni- cal ad- just- ment

In ordinary circumstances this is a reproducible vascularity-like prognostic phenotype. Here it is the counterexample the rest of the paper dismantles.

4.2 The phenotype is acquisition-dominated, not subtype-defined

Once acquisition variables enter, the reading changes. The raw cluster is associated with manufacturer at $p = 8.0 \times 10^{-104}$, and with neither molecular subtype ($p = 0.146$) nor recurrence ($p = 0.766$) (Table 6). The contingency is not subtle: Cluster 0 is 93% GE, Cluster 1 is 76% Siemens (Figure 6, Table 7).

Table 6: Association of raw cluster assignment with clinical and technical variables (χ^2 tests).

Variable	p-value	Reading
Manufacturer	8.0×10^{-104}	Strong association with cluster assignment
Molecular Sub-type	0.146	No significant association
Recurrence	0.766	No significant association

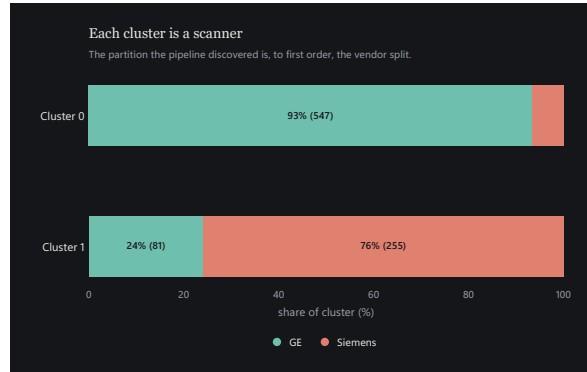
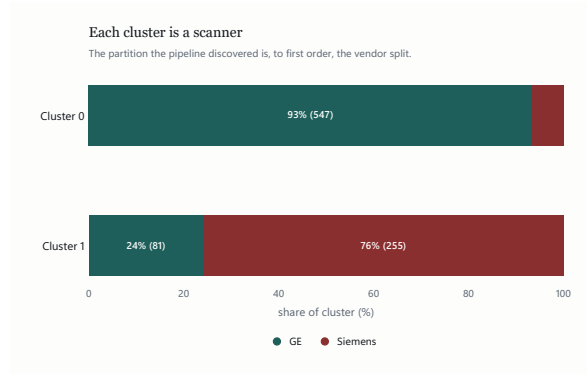
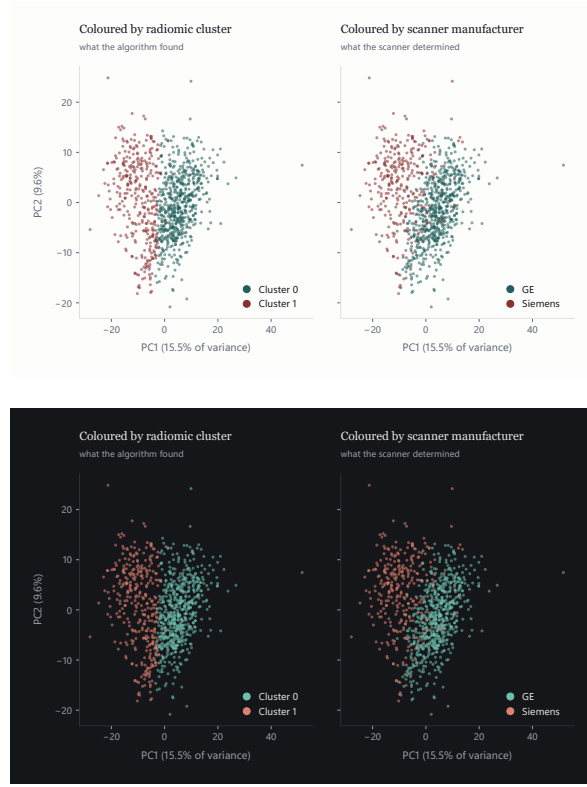


Figure 6: Manufacturer composition within each raw radiomic cluster.

Table 7: Raw cluster assignment by MRI manufacturer.

Cluster	GE	Siemens	Reading
Cluster 0	547	39	Predominantly GE
Cluster 1	81	255	Predominantly Siemens

The clearest way to see it is to recolour the same projection. Below is one PC1–PC2 scatter — the same 922 points in the same positions — coloured first by the cluster the algorithm discovered, then by the scanner that acquired the image. Switch between them:



(a)

Figure 7: Raw PCA space coloured by radiomic cluster and by manufacturer. The clustering algorithm partitioned the cohort by the machine that acquired the image.

Looking into PC1 directly makes the mechanism explicit. TR, TE, the acquisition matrix and manufacturer all track PC1 far more closely than molecular subtype does, which stays below 0.01 across the leading components (Figure 8, Table 8).



Figure 8: Share of PC1 variance explained by acquisition and biological variables.

Table 8: Acquisition-variable dominance on the leading radiomic axes (η^2 , share of component variance explained).

Variable	Kind	PC1	PC2	PC3
TR (repetition time)	Technical	0.755	0.567	0.421
TE (echo time)	Technical	0.632	0.355	0.198
Acquisition matrix	Technical	0.567	0.286	0.047
Manufacturer	Technical	0.489	0.119	0.011
Field strength	Technical	0.093	0.036	0.001
Slice thickness	Technical	0.075	0.130	0.018
Molecular subtype	Biological	0.005	0.010	0.035
Recurrence	Biological	0.002	0.005	0.022

This is the stability trap in its plainest form. The partition is not noisy — it is stable, because the scanner-domain structure it recovers is itself perfectly

η^2 is inflated for a continuous acquisition parameter when each distinct observed value forms its own group, as TR does here (337 distinct values). Grouped into 20 quantile bins TR still explains 0.56 of PC1, and a purely linear fit gives $R^2 = 0.34$. The ordering does not change: acquisition first, biology nowhere. The full sensitivity table is written to `_data/acquisition_eta2_sensitivity.csv`.

reproducible. High bootstrap ARI is evidence that the technical signal is consistent, not that the biology is real.

4.3 Harmonization preserves stability and destroys identity

Residualization and ComBat both remove the manufacturer association — and take the phenotype with them.

4.3.1 Raw

Measure	Value
Mean bootstrap ARI	0.893 (SD 0.084)
ARI vs raw labels	1.000
Manufacturer association	8.0×10^{-104}
Molecular subtype association	0.146
Recurrence association	0.766

The baseline. Stable, and defined by the scanner.

4.3.2 Manufacturer residualized

Measure	Value
Mean bootstrap ARI	0.919 (SD 0.047)
ARI vs raw labels	0.056
Manufacturer association	0.530
Molecular subtype association	0.717
Recurrence association	0.075

Manufacturer dependence is gone. So is the partition: an ARI of 0.056 against the raw labels means the two solutions agree at close to chance.

4.3.3 ComBat harmonized

Measure	Value
Mean bootstrap ARI	0.903 (SD 0.058)
ARI vs raw labels	0.065
Manufacturer association	0.177
Molecular subtype association	0.809
Recurrence association	0.056

The same story by a different route. Clusterability survives; identity does not.

The ARI between residualized and raw labels was 0.056; between ComBat and raw, 0.065. Both indicate

near-total loss of partition identity. What did *not* degrade is clusterability: both spaces still split into internally consistent two-cluster solutions, with mean bootstrap ARI of 0.919 and 0.903 respectively (Figure 9, Table 9).

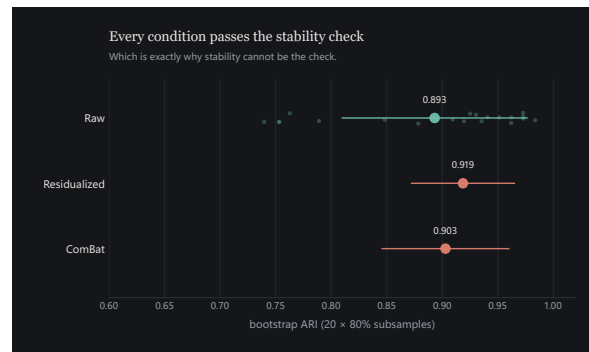
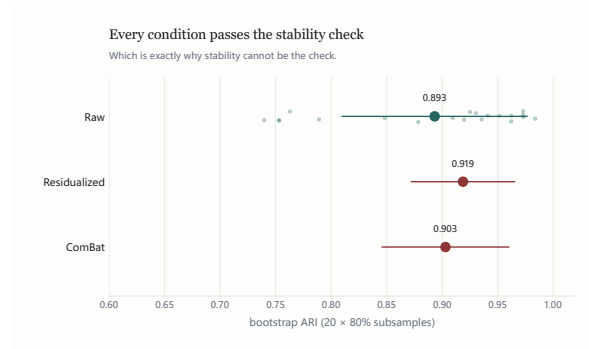


Figure 9: Bootstrap stability across raw and harmonized conditions. Every condition passes.

Figure 10 makes the same point geometrically: the same patients, in the same projection, sorted into a partition that bears almost no relation to the original one.

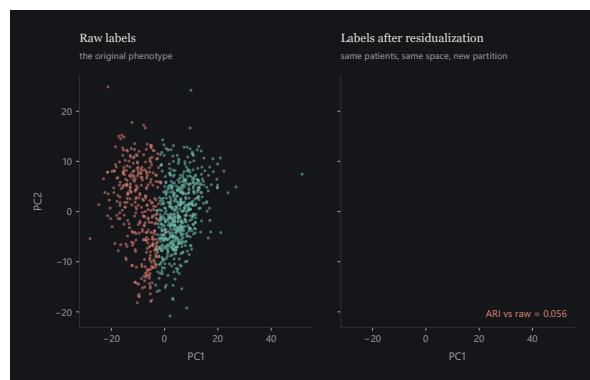


Figure 10: The same PCA space labelled by the raw partition and by the partition rediscovered after manufacturer residualization.

Plotting the two properties against each other separates concepts that are routinely conflated. Bootstrap stability asks whether a cluster solution can be *repeated* under resampling. Partition identity asks whether the *same patients* stay together after a technical adjustment. Here stability is preserved and identity is not — and a validation workflow that looks only at the first would certify this failure as a success.

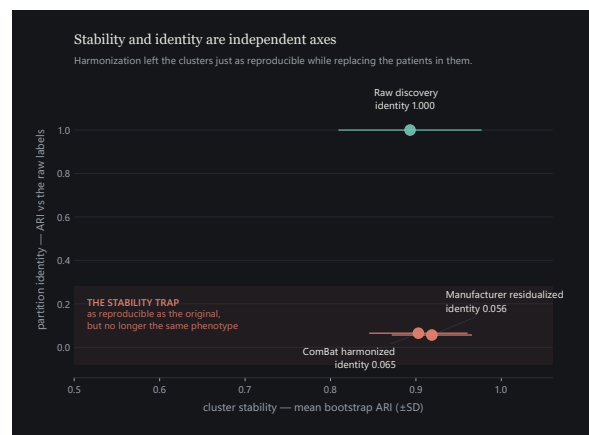
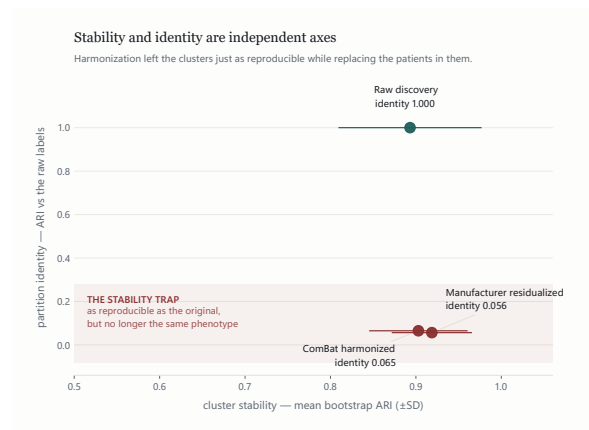


Figure 11: Stability and identity are independent axes. Harmonization moves a phenotype down the identity axis without moving it along the stability axis at all.

Table 9: Stability versus identity across raw and harmonized conditions.

Condi- tion	Mean boot- strap ARI	SD	ARI vs raw la- bels	Read- ing
Raw	0.893	0.084	1.000	Stable base- line parti- tion
Man- ufac- turer residu- alized	0.919	0.047	0.056	Sta- ble, but nearly unre- lated to the raw labels
Com- Bat harmo- nized	0.903	0.058	0.065	Sta- ble, but nearly unre- lated to the raw labels

4.4 Neither the survival signal nor the residual structure supports a bio-marker claim

The HR+/HER2+ signal does not survive adjustment. Entered into a Cox model alongside manufacturer, the cluster hazard ratio falls from 4.76 to 2.48 and loses significance (95% CI 0.46–13.39; $p = 0.291$). Manufacturer is likewise non-significant in this subgroup (HR 2.99; $p = 0.200$) — but with 10 events there are not enough degrees of freedom to separate two correlated predictors, and the honest reading is that the design cannot distinguish them, not that neither matters (Table 10).

Table 10: Manufacturer-adjusted Cox model in the HR+/HER2+ subgroup (10 events).

Co- vari- ate	Co- effi- cient	Haz- ard ratio	95% CI	p- value	Read- ing
Raw clus- ter	0.908	2.48	0.46– 13.39	0.291	At- tenu- ates and loses sig- nifi- cance
Man- ufac- turer: Siemens vs GE	1.097	2.99	0.56– 16.03	0.200	Non- sig- nifi- cant, but collinear with clus- ter

Structure does remain after the technical confounders are removed — but it is better described as a size-correlated residue than as an orthogonal discovery. The post-harmonization clusters are no longer detectably coupled to manufacturer, and associate most strongly with tumour bounding-box volume ($p \approx 10^{-26}$). Against the prespecified criteria, that residue fails on every count (Table 11): the recurrence association is weak, radiomic scores add nothing after clinical covariates, and the scores are not directionally consistent within vendors.

Table 11: Prespecified biomarker criteria applied to the post-harmonization residual structure.

Criterion	Observed re-sult	Status
Endpoint association	Cluster-level recurrence signals were weak and did not survive adjustment	✗ Not met
Incremental predictive value	Radiomic/vascularity-augmented models did not improve on clinical covariates	✗ Not met
Within-vendor consistency	Partition recovery differs sharply by vendor (ARI vs raw: 0.57 Siemens, 0.07 GE)	✗ Not met
Independence from tumour burden	Strongest surviving association is with bounding-box volume ($p \approx 10^{-26}$)	✗ Not met

The conclusion is not that radiomic clustering is useless. It is that this apparent discovery must be downgraded to a false positive once it passes through acquisition-aware falsification.

💡 The minimum a phenotype claim should have to report

1. **Multi-condition discovery.** Re-run discovery after residualization and after ComBat, not only on raw features.
2. **Partition identity, not just stability.** Report ARI of harmonized labels *against the original labels* — the number that collapsed here.
3. **Acquisition variance partitioning.** Report η^2 or equivalent for acquisition *and* biological variables on the leading components.
4. **Harmonization sensitivity of clinical claims.** Re-estimate every survival or outcome association after technical adjustment.
5. **Within-vendor sanity checks.** Repeat discovery inside each vendor stratum; a biological phenotype should reappear in both.

5 Discussion

5.1 The negative result is the result

A radiomic partition can be internally stable, visually coherent and apparently prognostic while failing to represent a biological subgroup. Our two-group partition looked compelling on every metric that is conventionally reported: it was reproducible under resampling, carried a vascularity-associated profile, and suggested an HR+/HER2+ subgroup with elevated recurrence. Under inspection it aligned with scanner manufacturer and pulse sequence.

This is what the term *stability trap* is for. The raw phenotype was not meaningless noise — it was stable precisely because the acquisition signal is stable. The bootstrap scores were misleading only because they were read as evidence of biological truth. The accurate and much narrower inference is that the clustering algorithm recovered a strong, reproducible signal present in the radiomic feature matrix. The falsification chain then establishes that the dominant signal is technical. If there is a positive finding here, it is the invalidation of the phenotype claim, not the identification of a new phenotype.

5.2 The danger is in the language

Feature-wise, the separating variables made sense. SER entropy, SER homogeneity, PE variability, enhancement-volume ratios and wash-in/washout descriptors all suggest increased or decreased vascularity and

perfusion. Enhancement-history descriptors are especially interpretable that way in breast DCE-MRI, where enhancement behaviour is already the central radiological finding.

That interpretability is what makes the trap dangerous: familiar vocabulary coats a technically induced partition. This is not an argument for avoiding biologically plausible language — enhancement features are not biologically meaningless. It is an argument that plausibility is not evidence. When the leading features are acquisition-sensitive, vascularity descriptors are a *hypothesis*, not a confirmation, and the profile should be stated as such — “these features are known to associate with vascularity, and the clusters separate on those features” — until the acquisition-aware checks are passed.

5.3 Stability and identity are separate validation targets

Harmonization is what makes the distinction visible. It did not destroy cluster stability; residualization and ComBat both produced perfectly stable two-cluster solutions. What changed is that the patients in those clusters were no longer the same patients.

In practice this means radiomic validation needs more than silhouette scores and bootstrap stability. A complete phenotype paper should report cluster preservation after acquisition correction, the technical-versus-biological dependencies of the leading components, the harmonization sensitivity of any clinical association, and the conformance of the post-harmonization structure to prespecified biomarker criteria.

5.4 Why within-vendor recovery differs

One asymmetry deserves comment. Repeating discovery inside each vendor stratum recovers the original partition much better within Siemens (ARI 0.574) than within GE (ARI 0.069). The cohort is GE-dominated (628 versus 294), so the raw Cluster 0 is essentially “the GE patients” and carries little internal structure to recover, whereas the smaller Siemens stratum retains a more distinctive feature profile. This is consistent with a partition built from vendor identity rather than biology, and it is also a reminder that within-vendor checks need to be interpreted against stratum size.

5.5 Limitations

The HR+/HER2+ survival analysis rests on 10 events. The divergence between subgroups is dramatic and the confidence interval is correspondingly wide; a larger

cohort would be needed to quantify any subtype-specific prognostic effect, and this constraint is endemic to medical imaging analysis.

k-means on principal components was legible and adequate for a falsification exercise, but the technical-confounding argument would be strengthened by repeating it under other unsupervised algorithms.

Harmonization is not ground truth. ComBat and residualization remove technical effects, but the post-harmonization subgroups are not thereby biological. Harmonization should be treated as an adversary in the falsification chain, not as an assurance of validity.

6 Conclusion

Radiomic phenotypes can be internally stable, visually coherent and apparently prognostic while failing multiple tests of biological identity. On a 922-patient multi-vendor breast-MRI cohort, a standard unsupervised workflow produced a stable, vascularity-like two-cluster solution that appeared to carry strong prognostic information in HR+/HER2+ disease. Acquisition-aware falsification changed the interpretation on every count: the leading components and the cluster labels were dominated by manufacturer and pulse-sequence structure; manufacturer adjustment removed the survival association; and residualization or ComBat replaced the original patient partition while leaving its internal stability untouched.

The broad claim is methodological. Stability is the first step of radiomic phenotype validation, not the last. Internal consistency should be accompanied by multi-condition discovery, harmonization sensitivity and within-vendor sanity checks. The analysis here is breast MRI, but the principle applies to any multi-centre or multi-vendor radiomics or imaging-AI setting in which technical variance is stable enough to be mistaken for biology.

7 Supplementary material

i S1 · Structural interpretation of the raw PCA and clustering output

Item	Interpretation
Scree plot	Variance declines progressively with no elbow; component count is a modelling choice
PC1–PC2 scatter	Patients form a continuous distribution, so any boundary is a gradient cut
Silhouette / CH at $k = 2$	Best on two of three metrics; Davies–Bouldin selects against $k = 2$
Bootstrap ARI 0.893	The partition is reproducible under resampling
Cluster–manufacturer χ^2	Reproducibility is inherited from the vendor split
Post-harmonization ARI	Stability is retained; identity is not

i S2 · Raw cluster assignment by molecular subtype and by recurrence

Table 12
By molecular subtype

Cluster	HER1+/HER2+ enriched	Luminal-like	Triple negative
Cluster 0	41	56	383
Cluster 1	18	48	212

By recurrence event

Cluster	No recurrence	Recurrence
Cluster 0	527	57
Cluster 1	306	30

Neither cross-tabulation shows the concentration that the manufacturer cross-tabulation shows. That contrast is the whole of Section 4.2 in two tables.

i S3 · Biological interpretation of the dominant feature patterns

Feature category	Cluster pattern	Interpretation offered by the language
SER en-tropy	Ele-vated in Cluster 1	Greater enhancement heterogeneity
SER ho-mogeneity	Ele-vated in Cluster 0	More spatially uniform enhancement
PE vari-ability	Ele-vated in Cluster 1	Less uniform peak enhancement
Wash-in / washout slopes	Dif-fer be-tween clusters	Perfusion-kinetic difference
En-hancement-volume ratios	Dif-fer be-tween clusters	Differing enhancing tissue fraction

Each row is a plausible sentence. None of them is evidence, because the same features are acquisition-sensitive (Figure 8).

i S4 · Within-vendor discovery

Table 13

Manufac-turer	n	ARI vs raw labels	Silhouette
Siemens	294	0.574	0.201
GE	628	0.069	0.140

i S5 · Notes on figures dropped from the source manuscript

Two supplementary figures in the original Word manuscript duplicated main-text figures — a PCA projection coloured by manufacturer and one coloured by raw cluster, both already shown in Figure 7 — and have been folded into the interactive toggle above rather than repeated. Supplementary Figure S1 (cluster distribution within each manufacturer) is the transpose of Figure 6 and is likewise not repeated.

Reproducibility

Every number, table and figure on this page is generated at render time from the pipeline’s own outputs — nothing is transcribed by hand.

Table 14: What produces what.

Component	Location	Role
Analysis pipeline	<code>src/radiomics_stats.py</code>	Preprocessing, PCA, k-means, stability, harmonization, survival
Export step	<code>_prep/export_data.py</code>	Ships the pipeline outputs into the CSVs this page reads
Figure theme	<code>_theme.py</code>	Palette, typography and light/dark rendering
Figure builders	<code>_figures.py</code>	Every figure on this page
Data layer	<code>_data/*.csv</code>	Patient-level frame, variance, η^2 , KM curves, result tables

Seeds and hyperparameters are fixed in the pipeline’s `config.py` (`RANDOM_SEED = 42`, 20 bootstrap iterations at 80% subsampling, `n_init = 20`). PCA variance ratios are recomputed here by SVD on the z-scored matrix and reproduce the pipeline’s values; the η^2 figures

reproduce the source manuscript’s Table 9 under the same grouping convention, with the sensitivity analysis noted in Section 4.2.

i Departures from the source Word manuscript

These are corrections, not reinterpretations. The analysis is unchanged.

- **Table 12 and heading 5.4** did not exist in the source; numbering here is continuous and generated.
- **Two supplementary figures** duplicated main-text figures and were dropped (S5 above).
- **SER** was defined inconsistently — “signal-enhancement ratio” in the results and “surface-enhancement-return” in the discussion. The first is correct and is used throughout.
- **The unadjusted Cox p-value** was reported as 0.02; the pipeline value is 0.025.
- **Zwanenburg et al.** was cited in text as 2020 but listed as the 2016 arXiv preprint; the published 2020 *Radiology* version is cited here.
- **Leithner et al. (2022)** was cited in text but listed under a co-author’s name; the reference is corrected to its first author.
- **A within-vendor asymmetry** (ARI 0.574 Siemens vs 0.069 GE) was present in the pipeline output but not discussed in the source; it is addressed in the discussion.

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